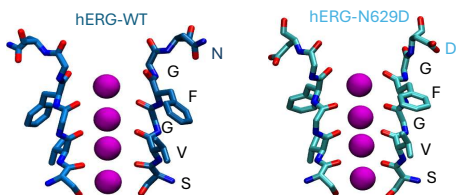


1. Introduction

Ion channels are key drug targets because they regulate essential physiological processes.¹ The hERG channel facilitates the passage of K⁺ ions and is crucial in cardiac repolarization – helping reset the heart's electrical state after each beat. Many drug candidates fail because they unintentionally block the hERG K⁺ channel, a major cause of drug-induced cardiotoxicity.²

To evaluate inhibitor effects, we first need to understand hERG's native function, including its low conductance and inward rectification,³ properties that remain elusive despite their importance in regulating heart activity.

Molecular Dynamics (MD) simulations of **wild-type (WT)** hERG fail to support stable K⁺ conduction due to force field (FF) limitations, SF instability, and reliance on non-physiological voltages (± 500 -750 mV). Recent studies show improved SF stability in the **N629D** mutant,⁴ and newly developed FFs enhance conductance in K⁺ channels.⁵

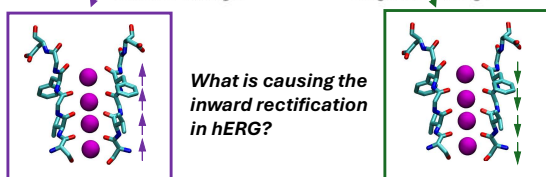
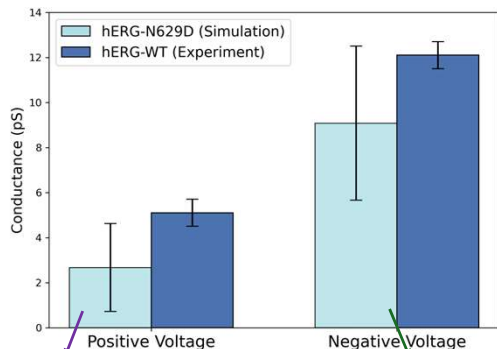


2. Research Aims

- Simulate hERG-WT and **N629D** systems using improved FFs.
- Achieve stable K⁺ permeation under near-physiological conditions using moderate voltage (± 300 mV).
- Understand the mechanism underlying inward rectification and low conductance of hERG.

3. Conductance Observed in hERG-N629D with Amber14sb-s3

Systems were simulated using the electronic continuum correction (ECC) optimized Amber14sb FF with 0.78 scaling, including additional scaling of the S3 carbonyl oxygen to enhance conductance.⁵ With this setup, the N629D mutant showed conductance within the experimental range for hERG-WT,⁶ and preserved inward rectification.

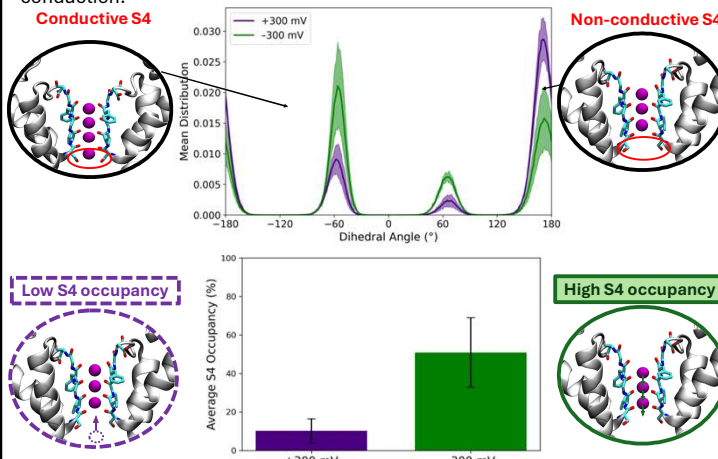


7. References

1. G. J. Kaczorowski, O. B. McManus, B. T. Priest and M. L. Garcia, *J. Gen. Physiol.*, **2008**, 131, 399–405.
2. T. M. Creanza, P. Delre, N. Ancona, G. Lentini, M. Saviano and G. F. Mangiatordi, *J. Chem. Inf. Model.*, **2022**, 62, 1170–1181.
3. D. A. Doyle, J. M. Cabral, R. A. Pfuetzner, A. Kuo, J. M. Gubis, S. L. Cohen, B. T. Chait and R. MacKinnon, *Science*, **1998**, 280, 69–77.
4. J. Li, R. Shen, B. Reddy, E. Perozo and B. Roux, *Sci. Adv.*, **2021**, 7, eab6203.
5. C. Hui, R. de Vries, W. Kopec and B. L. de Groot, *Proc. Natl. Acad. Sci. U.S.A.*, **2025**, 122, e2423866122.
6. A. Zou, M. E. Curran, M. T. Keating and M. C. Sanguinetti, *Am. J. Physiol. Heart Circ. Physiol.*, **1997**, 272, H1309–H1314.

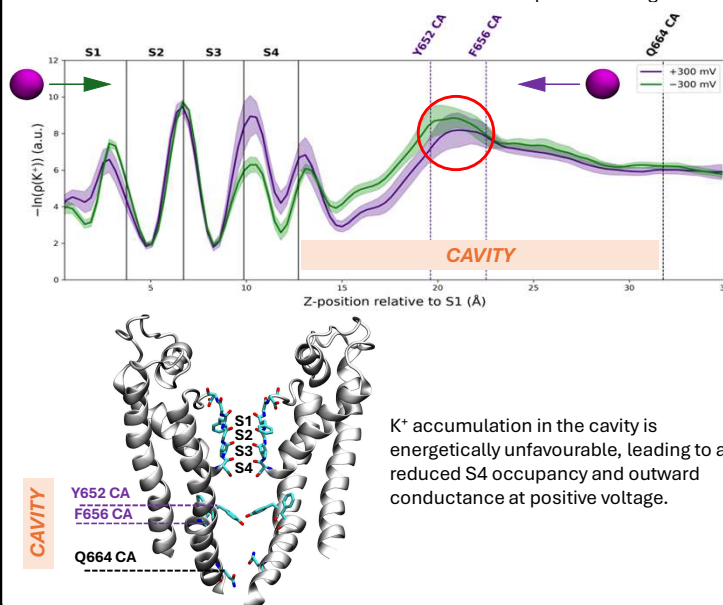
4. S4 Site as a Basis For Inward Rectification

We focused on the serine residue at the **S4 site**. Its sidechain shifts between conductive ($\pm 60^\circ$) and non-conductive ($\pm 180^\circ$) states depending on K⁺ binding to S4. However, S4 occupancy decreases at positive voltage, limiting outward conduction.



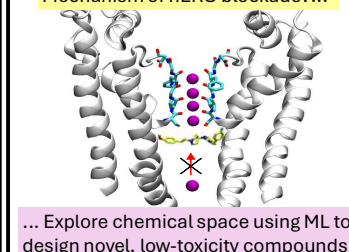
5. Cavity Restricts K⁺ Entry at Positive Voltages

Mean free energy profiles along the K⁺ conduction pathway reveal a **free energy barrier** between Y652 and F656 that limits K⁺ access to S4 at positive voltage.



7. What is next?

Mechanism of hERG blockade?...



6. Conclusions

- Observed K⁺ permeation under stable, near-physiological conditions using improved FF and moderate voltage.
- Improved FF yield conductance values consistent with experiment.
- Inward rectification in hERG arises from voltage-dependent S4 occupancy, where K⁺ access from below is impeded at positive voltage, reducing outward conduction.